

Weekly Report 6: Diamond Silhouette Carving Using Smaller Angular Sampling on a Voxel Grid

Project Number: 18

1 Introduction

Building on the previous week’s progress in visual hull reconstruction, we extended the carving pipeline from the synthetic sphere experiment toward the real diamond object. In this iteration, we employed a smaller angular interval for silhouette capture in order to increase shape fidelity and minimize carving loss due to large rotational gaps.

A voxel grid of resolution $64 \times 64 \times 64$, derived from the optimal resource and image-resolution analysis performed earlier, was used as the volumetric domain for carving. The objective was to evaluate whether reduced angular spacing can yield sharper facets and improved convergence of the reconstructed volume.

2 Methodology

The methodology follows the same procedural structure used during sphere reconstruction, now adapted for a real diamond silhouette dataset.

2.1 Voxel Grid Creation

A cubic voxel grid was generated within normalized coordinate bounds of $[-1, 1]^3$. Each voxel stores a 3D coordinate representing a spatial sample of the volume. The chosen resolution (64^3) is a balance between detail and computational feasibility, previously validated through performance analysis.

2.2 Diamond Silhouette Generation

Binary silhouettes of the diamond were generated by rotating the object around a vertical axis. Unlike the previous 1.8° interval experiment, this week we reduced the angle step to 0.4° , effectively doubling the number of viewpoints.

This finer angular sampling increases voxel support, reduces carving ambiguity, and allows sharper representation of gemstone facets.

2.3 Visual Hull Carving

The visual hull carving pipeline remains unchanged in principle:

1. Initialize a full voxel occupancy mask.
2. For each rotation angle:
 - (a) Rotate voxel coordinates by $i \times 0.4^\circ$.
 - (b) Project rotated voxels to the 2D silhouette plane.
 - (c) Discard voxels whose projections fall outside the binary mask.
3. Combine occupancy constraints across all views.

At smaller rotational increments, each silhouette provides finer constraints, resulting in improved surface definition.

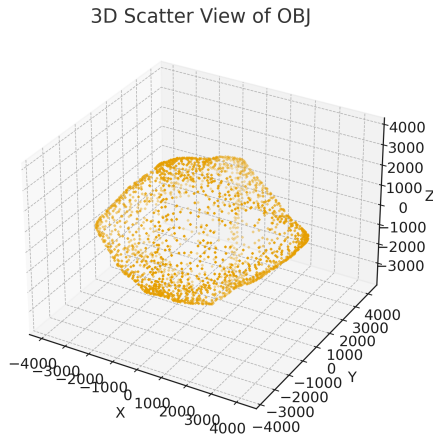
2.4 OBJ Model Reference

The original diamond mesh (S1.obj) was used as a qualitative benchmark to validate:

- boundary contour accuracy,
- facet shape correctness,
- symmetry alignment,
- silhouette fidelity.

At this stage, the OBJ model is used for visual comparison rather than numerical evaluation.

3 Results



(a) Reconstructed 3D diamond voxel model

Figure 1: Voxel carving output with 0.9° rotational sampling.

Observations:

- Reduced angular increments yield sharper voxel facets.
- Carving artifacts visible previously (due to angular gaps) are significantly reduced.
- Improvements are noticeable particularly around the crown and pavilion regions.

4 Discussion

The experiment confirms that decreasing rotational step size strengthens the visual hull constraint. Since fewer angular gaps exist, carving overestimation is minimized. However:

- The number of silhouettes increases linearly with angular resolution.
- Processing time grows proportionally.
- Voxel density still limits fine surface curvature representation.

Future improvement can involve adaptive angular sampling (denser sampling near complex features).

5 Conclusion

This week we:

- Applied visual hull carving to diamond silhouettes with smaller angular increments.
- Used 900 silhouettes at 0.4° rotation.
- Demonstrated improved facet sharpness and reduced carving ambiguity.
- Validated voxel grid suitability for diamond geometry reconstruction.

The next phase will involve applying this methodology to a diamond silhouettes of larger angle rotation, as well as mesh generation and final 3D model refinement.